

TABLE OF CONTENTS

	Page
TITLE PAGE	i
CERTIFICATE OF PANEL APPROVAL	ii
ABSTRACT	iii
DEDICATION	iv
ACKNOWLEDGMENT	v
TABLE OF CONTENTS	vii
LIST OF NOTATIONS	ix
LIST OF FIGURES	xii
CHAPTER	
1	
INTRODUCTION	1
1.1 Background of the Study	1
1.2 Statement of the Problem	4
1.3 Objectives of the Study	4
1.4 Methodology	5
1.5 Significance of the Study	6
2	
PRELIMINARIES	7
2.1 Strong Derivative	7
2.2 McShane and Henstock Integrals	17
2.3 Inner Variation	26
2.4 The Cantor set P	37
3	
EQUI-INTEGRABILITY AND EGOROFF'S THEOREM	47
3.1 Equi-integrability	47
3.2 Egoroff's Theorem	79
4	
UNIFORM DIFFERENTIABILITY	92
4.1 Uniformly Strongly Differentiable	92
4.2 Uniformly Differentiability	109
5	
LIPSCHITZ AND GENERALIZED AC FUNCTIONS	111
5.1 Lipschitz Condition	111

5.2	Lipschitz and Generalized AC Functions	118
6	HARNACK EXTENSIONS	144
6.1	Harnack Extension on Intervals in $\mathbb{R}$	144
6.2	Harnack Extension on Intervals in $\mathbb{R}^2$	182
7	SUMMARY AND RECOMMENDATIONS	190
7.1	Summary	190
7.2	Recommendations	219
	BIBLIOGRAPHY	220
	CURRICULUM VITAE	222
	CERTIFICATE OF AUTHENTIC AUTHORSHIP	224